Project Title

Global Aviation Disruption Analysis during the 2026 Iran–US Conflict

Scenario

You are working as a Data Analyst for an international aviation monitoring organization. The organization wants to understand how geopolitical conflicts disrupt global aviation operations.

Your task is to analyze the dataset and produce insights for airline companies, airport authorities, and policy makers.

Source Link :

https://www.kaggle.com/datasets/zkskhurram/global-civil-aviation-disruption2026-iranus-war/data

Phase 1 — Data Engineering & Data Understanding (20 min)

Students must first prepare the dataset.

Tasks

1.Load the dataset into Pandas.

2.Identify dataset structure.

Questions:

> What are the main entities represented in the dataset?

> Which columns represent time, location, airline operations, or disruptions?

> Are there missing or inconsistent values?

> If some columns contain text dates or numeric strings, how would you convert them?

Technical Requirements

Students must use:

> pandas

> numpy

Expected operations:

> missing value detection

> type conversion

> basic dataset profiling

Phase 2 — Exploratory Data Analytics (30 min)

Business Objective

Airlines want to understand how severe the disruption was and where it occurred most frequently.

Questions

> Which countries or regions experienced the highest aviation disruptions?

> Which airlines or airports were most affected?

> What time period shows the highest disruption activity?

> Are disruptions concentrated in specific flight corridors or airspace zones?

Required Visualizations

Students must create at least:

> Bar chart

> Time series chart

> Pie chart

> Heatmap

Libraries:

> Matplotlib

> Seaborn

Phase 3 — Advanced Visualization (30 min)

Business Objective

Airline executives want interactive insights to understand disruption patterns globally.

Students must use Plotly.

Tasks

Create the following:

> Interactive geographic visualization

Map showing where aviation disruptions occurred.

> Timeline visualization

When did disruptions peak?

> Airline impact dashboard

Which airlines experienced highest cancellations?

> Operational impact visualization

Compare disruptions across regions.

Expected tools:

> plotly.express

> plotly.graph_objects

Phase 4 — Probability & Risk Analysis (20 min)

Business Question

Airlines want to estimate risk levels during conflict periods.

Students must analyze:

1.What is the probability that a flight operating in a conflict region will be disrupted?

2.How does disruption probability differ between regions or airlines?

3.If a flight passes through multiple high-risk regions, how does the probability of disruption change?

Students should:

> calculate probabilities

> interpret operational risk

Phase 5 — Statistical Testing (20 min)

Students must choose appropriate statistical tests to answer the following business questions.

Question 1 — Operational Cost Analysis

Airlines suspect that flight delays increased significantly during the conflict period.

Students must test:

Is there a statistically significant difference in flight delays before vs during disruption periods?

(Expected statistical approach: t-test)

Question 2 — Regional Disruption Analysis

Airline regulators want to know:

Do different regions experience significantly different levels of aviation disruption?

(Expected statistical approach: ANOVA)

Question 3 — Airline Risk Analysis

Aviation authorities want to understand:

Is there a relationship between airline identity and disruption occurrence?

(Expected statistical approach: Chi-square test)

Phase 6 — Data Science Insight & Decision Making (20 min)

Students must produce executive-level insights.

Business Questions

> Which global air routes are most vulnerable during geopolitical conflicts?

> How should airlines re-route flights to minimize disruption risk?

> Which airports should be considered backup hubs during crises?

> What early signals in the dataset indicate potential aviation crises?

Skills Being Tested

Skill	Tested Through
Data Engineering	data cleaning & loading
Data Analytics	EDA questions
Data Science	statistical inference
Python	numpy & pandas
Visualization	seaborn, matplotlib, plotly
Probability	disruption risk estimation
Statistics	ANOVA, t-test, Chi-square

✅ This project forces students to:

think like aviation analysts

apply statistics without being told explicitly

combine analytics + visualization + inference

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Start coding or generate with AI.
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